Cell

Leading Edge
Review

Mechanochemical Feedback Loops in Development and Disease

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https://doi.org/10.1016/j.cell.2019.05.052

There is increasing evidence that both mechanical and biochemical signals play important roles in development and disease. The development of complex organisms, in particular, has been proposed to rely on the feedback between mechanical and biochemical patterning events. This feedback occurs at the molecular level via mechanosensation but can also arise as an emergent property of the system at the cellular and tissue level. In recent years, dynamic changes in tissue geometry, flow, rheology, and cell fate specification have emerged as key platforms of mechanochemical feedback loops in multiple processes. Here, we review recent experimental and theoretical advances in understanding how these feedbacks function in development and disease.

Introduction

The molecular and cellular processes mediating mechanical force generation, transmission, and sensing in developmental, homeostatic, and disease-related processes are increasingly well understood (Heisenberg and Bellaïche, 2013; Lecuit et al., 2011). Force generation at the cellular scale is typically achieved by the interaction of motor proteins, such as myosin, with filamentous elements, such as F-actin, thereby changing the architecture, rheology, and mechanical stress of the cell cytoskeleton. In addition, mechanical forces can be generated by polymerization-driven growth of cytoskeletal structures and cell volume regulation (Phillips et al., 2012). For cell-scale force generation to affect tissue morphogenesis, forces need to be transmitted between cells and their environment, which are typically other cells or the extracellular matrix (ECM). Such force transmission is achieved by specific adhesion proteins, such as cadherins at cell-cell contacts and integrins at cell-ECM contacts (Heisenberg and Bellaïche, 2013), but can also be mediated by unspecific friction (Bergert et al., 2015). Moreover, cells can passively respond to force application by deformation and translocation but can also actively respond by force-induced changes in biochemical signaling, a process termed "mechanosensation" (Petridou et al., 2017). Such mechanosensitive responses open the possibility for various forms of feedback loops where biochemical signals induced by mechanical signals change the ability of cells to generate, transmit, or receive forces, thereby modifying the generation and/or impact of the initial inducing mechanical signal. Feedback loops between mechanical and biochemical signals, also termed "mechanochemical feedback loops," have been described to function both at the cell and tissue scale and have been proposed to be essential for several key cellular processes in development and disease, such as cell migration, polarization, and fate specification (Chan et al., 2017; Goehring and Grill, 2013). Mechanochemical feedback loops have also caught the interest of physicists studying the fundamental differences between passive and active materials (Prost et al., 2015). Here, activity is defined at the microscopic scale as the steady-state conversion of chemical energy, in the form of ATP molecules, into mechanical work. Most of these biophysical studies have focused on understanding the rheological and emergent hydrodynamical properties of active materials (see Box 1 for glossary)—for instance, how mechanical forces generated by the activity of molecular motors shape the cell cytoskeleton (Heisenberg and Bellaïche, 2013; Prost et al., 2015). However, there is also an increasing number of biophysical studies, both experimental and theoretical, attempting to conceptualize the converse feedback on how large-scale mechanical properties of the cytoskeleton can impact on biochemical signaling. In this Review, we will focus on mechanochemical feedback loops, discussing what key ingredients are needed for them to emerge and how they function in different developmental, homeostatic, and disease-related processes.

Mechanochemical feedback loops in their simplest expression imply an effective back-and-forth between mechanical and biochemical signals. For instance, a mechanical signal would elicit a biochemical signal in the receiving cell, which in turn would modify the original mechanical stimulus, thereby establishing a positive or negative feedback. These mechanochemical feedbacks are often found on a subcellular scale, where forces produced by the cytoskeleton lead to rheological changes of the cytoskeleton itself that modulate its force-generating capacity and typically rely on tension-mediated changes in protein structure or rearrangement and/or flows of molecular components. As many of these mechanosensitive feedback processes have already been extensively covered by previous reviews (Petridou et al., 2017; Pinheiro and Bellaïche, 2018; Schiller and Fässler, 2013), we will only shortly summarize them at the beginning of our Review.

However, mechanochemical feedback loops can also arise as an emergent property of interactions among many system components at the mesoscopic or tissue scale, which requires

Box 1. Glossary of the Terms Used in the Review

rheology: study of the properties of materials under an applied force, e.g., the extent of their viscous or elastic behavior
advection and diffusion: two possible mechanisms of transport—advection refers to directed transport via a global fluid flow, while diffusion refers to random transport (e.g. of cells or molecules) in the absence of net flows
active matter: generic term used in physics for materials that are driven out of thermal equilibrium by constant microscopic input of energy (e.g., ATP consumption), giving rise to novel behavior
Turing instability: generic class of mechanisms proposed by Alan Turing for *de novo* pattern generation based on the diffusion and cross-reactions of several morphogen species
buckling instability: mechanical transition from a flat material to a curved or "buckled" state due to compressive forces acting on it, which constitutes an alternative mode of *de novo* patterning and symmetry breaking during development
curvotaxis: ability of cells and tissues to measure and respond to large-scale curvature
lateral inhibition: ability of cells to inhibit a given signal, fate, or activity in their neighboring cells

qualitatively different theoretical approaches. These constitute a fascinating frontier for quantitative biology in the coming years, and we will primarily focus in this Review on those more indirect mechanisms at the systems level. In particular, we will describe how physical processes (driving cell and tissue morphogenesis) and biochemical processes (leading to cell fate specification and tissue patterning) influence each other and how their mutual interaction triggers tissue self-organization in development. We also will discuss that a number of mechanochemical feedback loops found in developmental settings appear conserved in various disease-related processes.

Mechanochemical Feedback Loops Mediated by Protein Conformational and/or Organizational Changes

The classical view of mechanochemical feedback loops is that proteins can respond to mechanical forces by conformational changes that modify their biochemical properties and/or network of interactors, thereby establishing a feedback loop. One of the best studied examples are cell-ECM and cell-cell contacts (Figure 1). Cell-ECM contacts are typically mediated by focal adhesions (FAs), which have been demonstrated to grow in response to the force applied. This mechanosensitive process is mediated by specific components of FAs, such as the adhesion receptor Integrin and adaptor protein Talin, changing their conformation in response to the applied force (Yan et al., 2015). This results in increased binding of Integrin to Fibronectin and Talin to Vinculin, ultimately strengthening the coupling of FAs to both the ECM via Integrin and the actomyosin cytoskeleton via Talin and Vinculin (Atherton et al., 2016). Force-induced conformational changes of Integrin and Talin also cause FAs to grow, which in turn affects the distribution of forces within this junctional complex and thus the likelihood of FA components to undergo conformational changes. Therefore, the mutual interplay between FAs and mechanical force constitutes a feedback loop where forces applied to FAs modify their composition and size via protein conformational changes, which in turn alters their susceptibility to forces.

Similar to FAs, cadherin-mediated cell-cell contacts have also been shown to be mechanosensitive (Pinheiro and Bellaïche, 2018). Forces at cadherin adhesion sites trigger conformational changes of the adaptor protein α-catenin to reveal cryptic binding sites for vinculin (Yonemura et al., 2010), ultimately increasing the coupling of cadherin adhesion sites to the actomyosin cortex. Increased cytoskeletal anchoring of cadherin adhesion sites in turn promotes cadherin clustering and stability at cell-cell contact sites by reducing their turnover rate (Engl et al., 2014). These force-induced changes in the size and composition of cadherin adhesion complexes impact on force distribution within these complexes and also the force exposition of mechanosensitive adhesion adaptors therein. Collectively, the mechanosensitivity of both cell-ECM and cell-cell adhesion sites allows those sites to adapt contact size and composition to the magnitude of applied forces, thereby preserving contact integrity even when exposed to a wide range of forces.

There are also examples where force-induced protein conformational changes directly affect the force-producing processes. In the *Drosophila* embryo, for instance, anterior endoderm specification has been proposed to be triggered by compression forces from the global extension of the germband (Desprat et al., 2008). These compression forces are thought to elicit the expression of the transcription factor *twist* by molecular stretching of the highly conserved β-catenin-E-cadherin binding site, resulting in the junctional release of the co-transcriptional regulator β-catenin and, consequently, β-catenin-mediated transcription of target genes, including *twist*. Twist, in turn, controls the expression of upstream regulators of actomyosin contractility (Sandmann et al., 2007). Interestingly, mechanosensitive β-catenin activation has also been shown to be triggered by mechanical stresses similar to those elicited by growing tumors, driving oncogenic transformation of neighboring cells (Fernández-Sánchez et al., 2015). Given that transformed cells and tumors are known to change the mechanical properties of their microenvironment (Northcott et al., 2018), this points at a feedback loop where mechanical signals lead to oncogenic transformation of cells, and tumor growth affects cell and tissue mechanics.

Other examples, where force-induced protein conformational changes directly affect the force producing processes, involve mechanosensitive changes in the subcellular localization and conductivity of the non-selective cation channel Piezo (Figure 1). How mechanical signals modulate Piezo channel localization and function is not yet fully understood, but changes in its conformation are likely to be involved (Guo and MacKinnon, 2017). Piezo channels—and mechanosensitive ion channels in general—transform mechanical signals into biochemical signals by modifying intracellular ion concentration leading, for example,

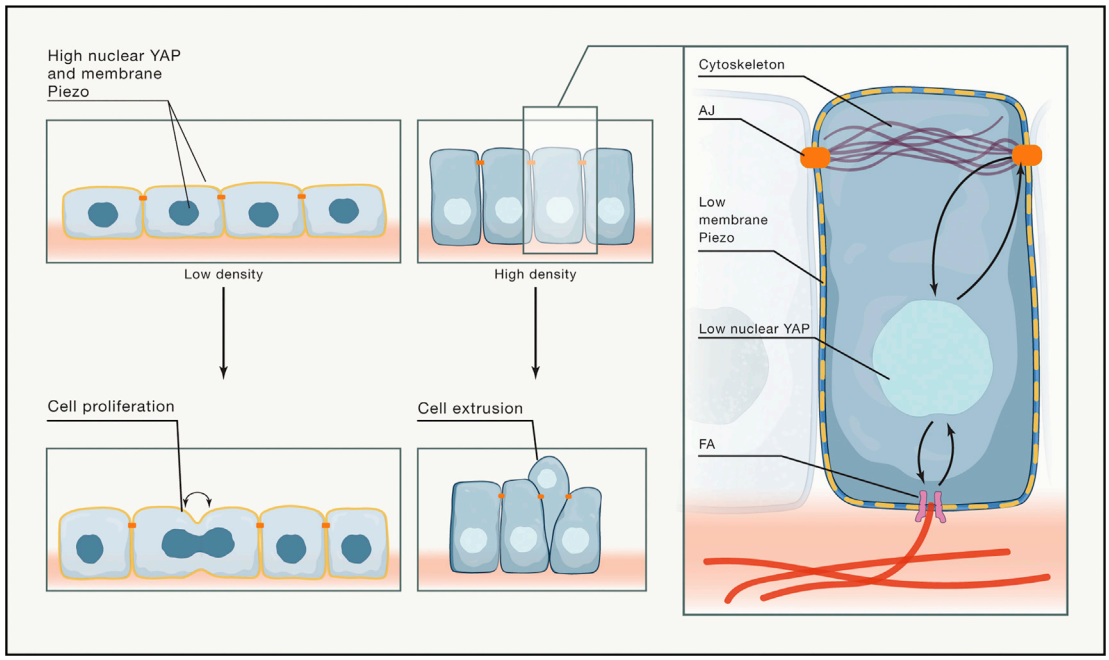


Figure 1. Mechanochemical Feedback Loops at the Subcellular Level

Schematics of the mechanisms of force generation and sensing in cells at the molecular level. Cells communicate biochemically and mechanically with their environment via cell-cell (adherens junctions [AJ]) or cell-matrix (focal adhesions [FA]) contacts. When stretched or at low cell density, cells undergo cell proliferation, and when compressed or at high cell density, cells are extruded, which together ensures tissue homeostasis. These mechanosensitive responses are mediated by cell compression and stretch, reducing and increasing the nuclear localization of the co-transcriptional activators YAP and/or TAZ (blue), respectively. Cell compression and stretch also control membrane localization and activation of mechanosensitive ion channels, such as the non-selective cation channel Piezo (orange).

to protein phosphorylation and thus activation. Interestingly, Piezo activation by cellular stretching at low cell density has been found to trigger intracellular Ca^{2+}-dependent ERK signaling, which in turn drives cell proliferation (Gudipaty et al., 2017). Once proliferation is underway, the primary mechanical stimulus of Piezo activation—low cell density and thus cellular stretching—will be progressively reduced, thereby providing a negative feedback signal on Piezo activation. Notably, modulation of Piezo activity by mechanical signals has also been implicated in live cell extrusion in overcrowded epithelia (Eisenhoffer et al., 2012), suggesting that Piezo mechanosensation plays a general role in maintaining tissue homeostasis. Interestingly, Piezo1, much like other mechanically gated ion channels, such as transient receptor potential (TRP) channels (Sidi et al., 2003), has been shown to be dysregulated in numerous solid and non-solid tumors (Northcott et al., 2018), supporting their role as a guardian of tissue homeostasis.

Finally, mechanosensation by the co-transcriptional activators YAP and TAZ has also been implicated in mediating the effect of cellular crowding and compression on cell proliferation, specification, and growth (Azzolin et al., 2014; Dupont et al., 2011). One recent example for such function is the specification of the micropyle precursor cell (MPC) within the follicular epithelium surrounding the oocyte during zebrafish oogenesis (Xia et al., 2019). Specification of a single MPC depends on the expression and nuclear translocation of TAZ. TAZ triggers massive growth of the MPC, which leads to mechanical compression and deformation of its neighboring cells and, consequently, depletion of nuclear TAZ in these cells, presumably due to the sensitivity of TAZ nuclear localization to mechanical force-induced conformational changes of the cell cytoskeleton and/or nucleus (Dupont et al., 2011; Elosegui-Artola et al., 2017). This lateral inhibition mechanism triggers a positive feedback loop where TAZ-dependent growth of the dominant MPC cell inhibits specification of the same fate in neighboring cells, thereby limiting their growth and, consequently, facilitating the growth of the dominant cell (Xia et al., 2019). Given the well-documented crosstalk between cell fate specification and cell proliferation (Chan et al., 2017), there will likely be many more of such examples linking fate specification and morphogenesis via mechanochemical feedback loops at the tissue-scale (see also below).

Mechanochemical Feedback Loops Mediated by Cytoskeletal and Tissue Flows

Cytoskeletal flows are one of the simplest manifestations of activity in out-of-equilibrium materials: spatial gradients in either motor contractility or cytoskeletal polymerization cause flows, which have been studied in various key biological processes such as cell division, migration, and polarization. Although cytoskeletal flows can arise from purely pre-patterned cues—for instance, graded distribution of molecules controlling actin polymerization and/or actomyosin contraction—the advection they create provides a simple feedback for self-organization. For instance, in case activators and inhibitors of contractility

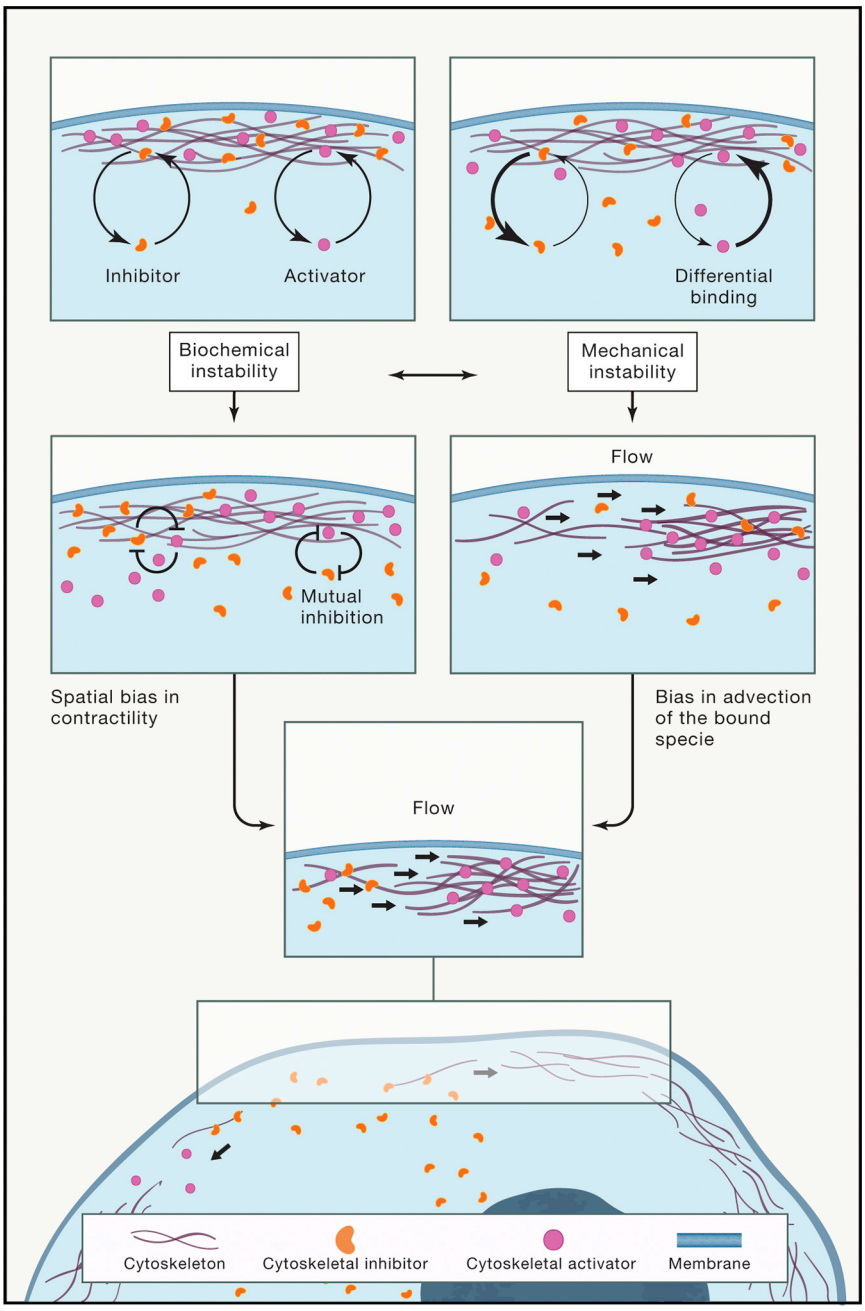


Figure 2. Mechanochemical Feedback Loops via Cytoskeletal Flows

Schematics of the mutual feedback between cytoskeletal mechanics and biochemical interactions. Activators (purple) and inhibitors (orange) of cytoskeletal polymerization and/or contractility can both cross-react with each other (e.g., via mutual inhibition), shuttle between a cortical and cytoplasmic pool, and be advected by active cytoskeletal movements (Goehring and Grill, 2013). Patterning from an initially homogeneous state can thus occur via two complementary feedback loops:

(1, left side) Activators and inhibitors can cross-react (e.g., via mutual inhibition), which breaks spatial symmetry via a biochemical reaction-diffusion instability. This creates spatial patterns of concentrations, which feed back on the cytoskeleton by creating spatial density changes in network density and/or contractility, resulting in directed network flows.

(2, right side) Spontaneous mechanical instabilities can arise from an initially uniform cortex due to self-reinforced contractile flows. If activators and inhibitors bind differentially to cytoskeletal elements, such flows will differentially advect them, biasing their spatial distribution and thus further affecting network contractility and/or polymerization.

Both of these feedback loops can interact with each other to give rise to spontaneous polarization of a cell (bottom center panel), as seen in *C. elegans* zygotes or migrating cells.

differentially bind to the cytoskeleton, they will be differentially concentrated by flows so that a resulting activator or inhibitor gradient can engage in a feedback loop with advection (Figure 2). These ideas have inspired novel physics on out-of-equilibrium descriptions of active materials, which provided quantitative insights on the conditions required for such mechanochemical instabilities to occur *in vivo* and *in vitro* (Bois et al., 2011; Hannezo et al., 2015; Mayer et al., 2010; Recho et al., 2013). Above a critical threshold of contractility, an initially homogeneous layer of actomyosin can be shown theoretically to undergo a phase transition toward a spatially patterned state that is maintained by steady flows (Figure 2). One of the best-studied examples of such actomyosin flow generation are migrating cells. There, actin polymerization at the leading edge is thought to trigger retrograde flows of the actin cytoskeleton. These retrograde actin network flows are resisted by friction to the ECM due to cell-ECM adhesion receptors, such as Integrins, which function as molecular clutches linking the actin network to the ECM, and eventually generate the force required for the migrating cell to move relative to the surrounding matrix (Barnhart et al., 2015; Liu et al., 2015; Maiuri et al., 2015; Recho et al., 2013; Ruprecht et al., 2015). Interestingly, retrograde actin flows in migrating cells not only directly trigger cell migration, but also enhance cell polarization along the direction of motion by flow-induced changes in the localization of biochemical signals and cellular compartments (Lavi et al., 2016). Likewise, actomyosin flows also affect its own network organization by changing the adhesion strength and/or orientation of actin filaments and myosin motors, as well as steering cell directionality at the rear end (Allen et al., 2018).

The morphogenetic function of actomyosin flows has also been extensively characterized in the *C. elegans* zygote, where flows of the actomyosin cortex polarize the zygote along its anteroposterior axis (Mayer et al., 2010). Anteriorly directed

actomyosin flows are initiated by a local disruption of the cortex at the posterior pole by the sperm centrosome, thereby generating a contractility gradient along the anteroposterior axis. Once initiated, the flows further polarize the zygote by transporting PAR proteins involved in anterior polarization, such as PAR3 and 6 (Goehring et al., 2011), to the anterior end of the zygote. Notably, not only polarizing proteins but also regulators of actin polymerization and actomyosin contractility, such as Cdc42, are anteriorly transported by the actomyosin flows (Motegi and Sugimoto, 2006), thereby generating a positive feedback and amplifying an initial local symmetry breaking event into embryo-scale polarization (Goehring and Grill, 2013; Gross et al., 2018). This system has been proposed to reside below a critical point at which spontaneous polarization occurs without guiding cues, but above a threshold at which amplification can occur (Gross et al., 2018).

More generally, such self-enhancing feedback loops could allow cells to rapidly and robustly respond to external cues. Indeed, sensitivity to external perturbation is an advantage generically conferred to self-organizing biological systems, and guided self-organized actomyosin flows have also been uncovered in a variety of other settings during vertebrate and invertebrate embryogenesis. For instance, the oscillatory actomyosin contractions at the apex of blastoderm cells during *Drosophila* germband extension have been proposed to rely on the ability of actomyosin flows to advect their activators and inhibitors—such as Rho1, Rok, and Mbs—in traveling pulsatile waves (Kumar et al., 2014; Munjal et al., 2015). The direction of these flows can be biased by a spatially heterogeneous distribution of E-cadherin at cell-cell junctions, ensuring polarized junctional remodeling (Levayer and Lecuit, 2013). Likewise, during *Drosophila* notum morphogenesis, junctional remodeling upon cellular division is achieved by self-organized actomyosin flows, the direction of which is determined by local E-cadherin dilution (Pinheiro et al., 2017). Finally, in zebrafish oocytes, self-organized actomyosin flows biased toward the animal pole have recently been found to trigger the segregation of ooplasm from yolk granules, thereby promoting oocyte polarization (Shamipour et al., 2019).

Self-amplifying loops via advection are not restricted to actomyosin flows but have also been proposed to operate at the tissue level. In the classical Murray-Oster mechanochemical theory, patterned bone condensations are driven by motile cells on collagen, which migrate toward zones of highest density in adhesive molecules (Murray et al., 1988). Given that cell migration deforms the collagen substrate, substrates in regions of high adhesion density will be compressed by the influx of migrating cells. Matrix compression then further increases the density of adhesion molecules and provides a positive feedback loop that ultimately amplifies cell condensation. From a mathematical perspective, there is a close relationship between those types of tissue-level "advective" instabilities and aforementioned self-reinforced actomyosin flows, which are broadly known as Keller-Segel models, initially developed to understand chemotactic patterns in bacteria (Keller and Segel, 1971). At the tissue scale, however, direct evidence for this type of mechanism remains scarce, possibly owing to technical limitations in imaging and precisely manipulating such tissue-wide processes. One recent example where such model could apply is hair-follicle formation in mouse, which originally had been proposed to solely arise from a Turing instability (Sick et al., 2006). There, self-amplifying cell condensations in the dermal cell layer have been found to create local differential stresses between the dermis and the overlying epidermal cell layer. This in turn triggers nuclear β-catenin translocation in epidermal cells above these condensations, potentially as a result of cell shape changes, thereby initiating localized hair follicle specification and differentiation in those cells (Shyer et al., 2017). In the next section, we will explore the interaction and feedback between the mechanical processes leading to cell and tissue morphogenesis and the signaling pathways and gene regulatory networks determining cell fate specification and differentiation.

Mechanochemical Feedback Loops Involving Cell Fate Specification and Differentiation

Different models have been brought forward to conceptualize the crosstalk between embryo patterning and morphogenesis. Broadly speaking, these models have historically fallen into two distinct categories, exploring either the chemical or mechanical basis of development. The initial concept of chemical descriptions of pattern formation came from Turing's seminal work (Turing, 1952), where he proposed that the cross-reactions between two differentially diffusing species, coined as "morphogens," could generically give rise to large-scale spatial-temporal patterns. This self-organized "chemical" model has been highly influential, although unambiguous experimental evidence for it has only more recently emerged—e.g., in patterning of the fish coat (Kondo et al., 2009) or mouse digit (Raspopovic et al., 2014). In parallel, pioneering work by D'Arcy Thompson (Thompson, 1942) initiated a separate effort to understand theoretically how mechanical forces can be used to understand the self-organized properties of animal development and shape. This self-organized "mechanical" model has in the meantime been corroborated by a large number of experimental studies on the mechanical basis of various morphogenetic processes in development (Savin et al., 2011). However, while the idea of a framework coupling chemical and mechanical processes dates back to the Murray-Oster mechanochemical theory 40 years ago (Murray et al., 1988), the theoretical foundations to couple these processes in a generic way are still being developed. Moreover, clear experimental evidence supporting the role of mechanochemical feedback loops in linking patterning to morphogenesis has only started to emerge in the past few years.

Role of Cell and Tissue Geometry

A number of recent studies have indicated a feedback between tissue geometry and patterning. For growing epithelial layers, highly generic mechanical instabilities are predicted to arise from differential growth. Indeed, if an initially flat epithelial sheet grows at a different rate compared to its surroundings—for instance, the underlying mesenchymal tissue or ECM—there will a critical amount of growth mismatch that will cause the epithelium layer to "buckle" out of plane into more intricate structures. This has been proposed in a variety of contexts, ranging from lung branching (Varner et al., 2015) to brain convolution (Tallinen et al., 2016) and gut shape acquisition (Hannezo et al., 2011; Savin et al., 2011; Shyer et al., 2013). Alternative

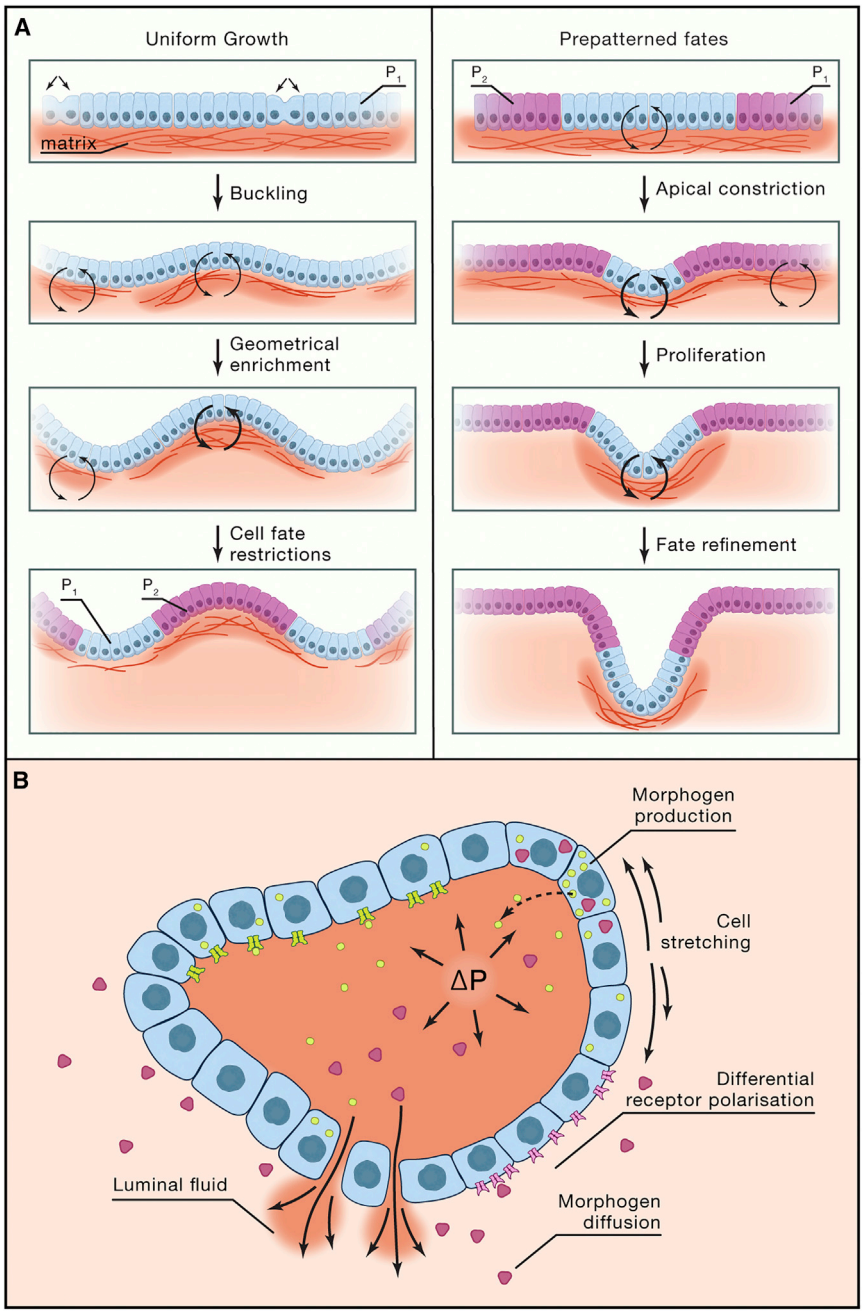


Figure 3. Mechanochemical Feedback Loops via Tissue Geometry

(A) Symmetry breaking of a growing epithelial sheet on an ECM substrate, which can happen via two complementary feedback loops:

Left (changes in tissue morphogenesis precede changes in cell fate specification): an initially flat layer of identical progenitors P_1, such as the mouse intestinal epithelium undergoing vilification, or lung and kidney epithelium undergoing branching morphogenesis, can undergo spontaneous symmetry breaking—a so-called buckling instability—when experiencing differential growth or stresses compared to its surroundings (orange mesh; e.g., mesenchymal tissues, ECM, smooth muscle layers), causing three-dimensional tissue folding. Such deformations result in changes in tissue curvature, which can be actively sensed within the epithelium (e.g., via YAP and/or TAZ nuclear localization) or cause local modulation in the concentration of diffusing biochemical species (e.g., Shh, BMP, FGF; drawn in shades of orange) from passive geometrical effects. This results in a feedback loop inducing different cell fates (P_1 and P_2) in different regions of the epithelium, which themselves can subsequently affect local force production and further morphogenetic events.

Right (changes in cell fate specification precede changes in tissue morphogenesis): in a layer containing pre-patterned domains of distinct cell types (P_1 and P_2, for instance during ventral furrow formation in *Drosophila* gastrulation), both cell types will typically have different mechanical properties. Higher apical contractility and/or larger division rates in the P_1 cells will cause local tissue bending, which can then be further refined via feedbacks on cell fate specification as outlined in the left panel.

(B) Interplay between luminal expansion, epithelial mechanics, and morphogen diffusion. The formation of luminal cavities during embryo development can provide a signaling center for diffusive biochemical signals, which can feed back on tissue contractility and thus subsequent lumen positioning and expansion but also provide mechanical signals via fluid pressure (ΔP). Epithelial tissues are typically polarized along their apicobasal axis, resulting in differential positioning of morphogen receptors to either the apical or basolateral areas. The ratio of apico-basolateral cell areas in turn depends on force balance within the epithelium, allowing for differential feedback loops to emerge as a result of changes in tissue and/or lumen geometry. For instance, apically secreted morphogens cannot activate basolateral receptors unless they diffuse through the epithelium, which could occur via growth-induced leakages or morphogenetically specified gaps in the epithelium. Furthermore, different cell types contribute in different ways to ion pumping and morphogen production, providing an additional feedback from cell fate specification to cell and tissue mechanics via tissue geometry.

mechanisms to sculpt epithelial layers in complex 3D shapes include differential constriction of cells along their apicobasal axis—for instance, during *Drosophila* gastrulation and vertebrate neural tube formation (Heisenberg and Bellaïche, 2013).

Interestingly, epithelial layers are typically polarized along their apicobasal axis, so that they can distinguish "top" versus "bottom" in a flat configuration. Consequently, these tissues are expected theoretically to be able to distinguish between regions of positive and negative curvatures, which translate in different ratios of apical versus basal area. Such properties suggest a potential feedback loop, where mechanical properties of an initially flat layer of cells would determine the three-dimensional (3D) structure it takes during buckling (breaking symmetry via a purely mechanical instability), and in turn, the acquired curvature of this 3D structure would feed back on local cellular identities and mechanical properties within the sheet (see Figure 3A for a schematic of this process). Experimental evidence for such curvature-sensing mechanism has recently been obtained by *in vitro*

studies on sheets of human epidermal stem cells cultured on sinusoidally patterned substrates (Walko et al., 2017). There, localization of the transcriptional co-activator YAP is restricted to cells at the tip, while cells in the valleys take on a more mesenchymal character. Activation of YAP has not only been associated with changes in cell proliferation and specification, but also with the generation of actomyosin-mediated mechanical forces (Porazinski et al., 2015). This points at the possibility of a feedback loop where out-of-plane tissue buckling triggers localized YAP activation, and YAP activation modulates actomyosin force generation within the cell sheet and, consequently, tissue buckling. However, whether these features are generic and how such ''curvotaxis'' emerges from a cellular and molecular perspective remain to be explored.

Related observations have also been obtained by *in vivo* studies on gut morphogenesis. Here, the differences in stresses and growth rates between the mesenchymal and epithelial cell layers and the surrounding contractile smooth muscle layers causes a buckling instability from an initially flat sheet to periodic 3D ridges—the so-called ''villi-intervilli structure'' (Shyer et al., 2013). Furthermore, although all epithelial cell initially secretes similar amount of the morphogen Sonic hedgehog (Shh) into the underlying mesenchyme, 3D tissue folding eventually will lead to a local enrichment of Shh within the mesenchyme positioned below the villi tips (Shyer et al., 2015). This local enrichment of Shh at the tips has been explained by purely geometrical arguments—i.e., the higher ratio of secreting epithelial cells to receiving mesenchymal cells in the tip compared to the bottom or valley region. Elevated Shh signaling within the tip mesenchyme then leads to increased expression of a second morphogen, bone morphogenetic protein (BMP), which in turn signals to the villi tip epithelium to inhibit stem cell fate and proliferation, thereby ensuring that stem cell production is restricted to the bottom regions of the curved epithelium (Shyer et al., 2015). Together, these findings suggest that a mechanical cue causing tissue buckling is effectively translated into a biochemical signal determining the spatial pattern of cell fate specification and proliferation (Guiu et al., 2019), thereby feeding back on the mechanics causing tissue buckling (Figure 3A). Notably, although these observations support the notion of a feedback between gut geometry, mechanics, and patterning, previous studies in mouse have argued that gut vilification in mouse is in fact mediated by biochemical Turing instabilities dependent on BMP signaling (Walton et al., 2016). This underlines the need for further studies specifically testing the respective influence of biochemical versus mechanical patterning cues on gut development in different model systems.

A similar interplay between tissue geometry and reaction-diffusion has been proposed to be at the core of branching morphogenesis (Menshykau et al., 2019). In mouse lung and kidney, the early patterns of branching observed *in vivo* can be accurately recapitulated by a Turing-like model of ligands and receptors, coupling the growth of the epithelial tree and its surrounding mesenchymal niche (Menshykau et al., 2019). In this system, the key requirement of a Turing instability—i.e., large differences between the diffusion rates of the activator and inhibitor species—is implemented by the fact that ligands are highly mobile in the mesenchyme, while receptors are largely immobilized in the epithelium. Robustness in the branching pattern emerges from coupling the local geometry of growing branches with production of morphogens (Menshykau et al., 2019). Interestingly, the mechanics of tip branching in the avian lung has also been shown to depend on the local patterns of differentiation of contractile smooth muscle, which locally constrict the epithelium to shape branching (Kim et al., 2013). This closely resembles the mechanisms discussed above on gut morphogenesis (Figure 3a), involving the presence of contractile and growth forces that give branches and tips their geometry, while morphogen reaction diffusion sets up the pattern of cell fate specification and proliferation. Nevertheless, the extent and directionality of the feedbacks between each of these processes remains to be demonstrated and quantitatively elucidated.

Finally, the formation of intercellular lumina has been shown to function as an effector mechanism mediating the crosstalk between tissue geometry and cell fate specification (Figure 3B). This can be achieved by the lumen either (1) acting as a signaling center or (2) triggering mechanical stresses in the lumen-lining cells due to pressurized fluid expansion—two possibilities we will successively discuss.

In zebrafish lateral line formation, periodic rosette formation is initiated by the coordinated apical constriction of small groups of cells under the influence of fibroblast growth factor (FGF) signaling (Durdu et al., 2014). This coordinated morphogenetic movement results in the formation of shared microlumen cavities, which act as signaling centers by locally concentrating FGF, thereby providing a positive feedback loop that triggers robust mechanosensory organ formation. The role of lumen signaling as an effector mechanism mediating the crosstalk between tissue geometry and cell fate specification might further be modified by the differential localization of morphogen receptors along the apicobasal axis of lumen-lining cells. Human embryonic stem cells, for instance, have been shown to self-organize by differential localization of morphogen receptors along their apicobasal axis (Etoc et al., 2016; Simunovic et al., 2018). This might be achieved by these cells differentially responding to signal-containing lumina depending on the size of their apical and basal surfaces and thus the localization of critical receptors relative to those lumina. Similarly, a recent study in mouse (Zhang et al., 2018) has proposed that the polarized basolateral localization of BMP receptors in epiblast cells, together with the spatially restricted diffusion of BMP4 ligands along the apicobasal extent of the epiblast cell layer, play important roles in embryo patterning (Figure 3B). However, whether in such cases polarized receptor localization can also feed back on tissue geometry still needs to be determined. Recent advances in the generation of *in vitro* mouse gastruloids, which recapitulate the self-organized symmetry breaking properties of the embryo along all three body axes (Beccari et al., 2018), will be key to understanding the biophysical basis of this interplay between mechanics, reaction-diffusion and tissue geometry.

Lumen formation not only relies on osmotic effects, cytoskeletal-mediated forces, and cell polarity (Bedzhov and Zernicka-Goetz, 2014; Chan et al., 2018; Dumortier et al., 2019), but also mechanically affects morphogenesis and cell fate specification in the surrounding cells (Figure 3B). In lung morphogenesis, for instance, luminal pressure has long been thought to affect

branching, and recent studies with bioengineered microfluidic cavities have shown that lung growth can be globally coordinated by the hydrostatic pressure in the lumen, determining the rate and timing of branching events (Nelson et al., 2017). Moreover, normal lung alveolar morphogenesis, which requires the correct specification of different alveolar cell types, depends on the mutual feedback between luminal pressure and cell-cell contact strength (Li et al., 2018). Increased luminal pressure, for instance during breathing, causes alveolar cell stretching, with some cells stretching more than others due to differences in their ability to reinforce their apical junctions in a myosin II-dependent manner (Li et al., 2018). The degree of cell stretching, in turn, determines which alveolar fate these cells will acquire. Whether this differential cell stretching occurs via divergence from initially small differences in contact strength between cells or as a more active mechanochemical feedback loop between fate specification and contact strength remains to be elucidated. Evidence for embryo geometry (Nishioka et al., 2009) and lumen formation influencing cell fate specification has also come from studies in mouse, showing that changes in luminal size in the early embryo correlate with cellular fate changes decreasing the ratio of inner cell mass to extra-embryonic trophectoderm cells (Chan et al., 2018). Finally, stresses arising from tumor growth can cause compression of blood vessels and elevate interstitial fluid pressure (Northcott et al., 2018), thereby influencing advective fluid flows and modifying the distribution of diffusing factors and metabolites within the tissue. This will change shear stresses sensed by tumor cells, which again might affect their adhesive properties and proliferation and/or death rates (Northcott et al., 2018). Cardiovascular homeostasis and associated diseases have also recently been investigated by modeling the mechanochemical couplings between reaction diffusion, flows, and cell mechanics (Buyandelger et al., 2014; Kim and Wagenseil, 2015). Likewise, the interplay between cellular mechanosensing, reaction diffusion, and mechanics is being actively investigated to understand bone repair and homeostasis (Borgiani et al., 2017), suggesting that both negative and positive mechanochemical feedback loops play also important roles in pathophysiological processes.

Collectively, these findings in multiple settings suggest that there are mutual feedback loops between tissue mechanics and patterning that can be generically mediated and understood by simple mesoscopic tissue properties, such as tissue geometry and luminal positioning. While in most of the described examples, clear evidence is provided for morphogenesis affecting cell fate specification, the converse effect of cell fate specification on tissue morphogenesis is often less clear. One case where such feedback has been more firmly established is the specification of anterior axial mesendoderm cells during zebrafish gastrulation that depends on high doses of Nodal signaling (Barone et al., 2017). In this process, Nodal signaling has been suggested to promote contact formation between mesendoderm cells, and, vice versa, contact formation has been shown to enhance Nodal signaling, thereby providing a positive feedback loop between contact formation and cell fate specification (Barone et al., 2017). However, whether generic design principles exist to understand the properties emerging from such couplings, such as robustness to fluctuations and initial conditions, scaling or evolvability remains to be explored. Recent theoretical efforts in this direction include studies on the interplay between reaction-diffusion instability and tissue mechanics to explore the hypotheses that morphogen concentration locally modifies cell shape or volume (Brinkmann et al., 2018; Dasbiswas et al., 2018; Okuda et al., 2018; Recho et al., 2019). This leads to theoretical paradigms bearing some resemblance to the generalized active gel descriptions mentioned above. Cell shape regulation can indeed feed back theoretically on morphogen response via direct mechanosensation, or by changing the ratio of apical to basolateral surfaces, assuming that receptor localization is polarized along the apicobasal cell axis (Dasbiswas et al., 2018). Cell volume changes can also theoretically provide feedbacks by modifying the local interstitial fluid pressure and thus morphogen transport (Recho et al., 2019). Interestingly, these models uncover specific parameter spaces where additional robustness is provided from such feedbacks, which now need to be systematically tested by experimental manipulations and quantitative studies of noisiness during morphogenesis (Haas et al., 2018).

Role of Cell and Tissue Rheology

Stem cell differentiation has been shown to be both influenced by rheological properties of its surrounding environment, such as substrate rigidity (Engler et al., 2006), as well as accompanied by chromatin condensation and marked changes in nuclear rheology (Heo et al., 2016a; Pagliara et al., 2014). This points to the intriguing possibility that mechanical forces control cell fate specification through their effect on nuclear rheology. Indeed, the nucleus has been found to function as a mechanochemical signaling platform able to respond to both cell-intrinsic and extrinsic forces transmitted by cytoskeletal elements (Cho et al., 2017; Kirby and Lammerding, 2018; Le et al., 2016). This function of the nucleus is thought to depend on the mechanosensitive nature of the nuclear lamina and its coupling to the actomyosin cytoskeleton via LINC proteins. Lamin A, for instance, a key component of the nuclear lamina, becomes stabilized under mechanical load, thereby increasing nuclear stiffness (Swift et al., 2013). Moreover, nuclear actin appears to be important by both propagating forces in its polymeric form (Feric and Brangwynne, 2013) and influencing transcription in its monomeric form (Treisman, 2013). Importantly, defects in nuclear mechanosensation have been associated with a wide array of pathologies, including cardiomyopathies and premature aging (Cho et al., 2017). Yet, how nuclear mechanosensing affects gene transcription and, consequently, cell fate specification, has only began to be unraveled.

One effector mechanism by which mechanical forces upregulate gene transcription on shorter timescales might be chromatin stretching caused by external physical forces and transmitted to the nucleus via the actomyosin cytoskeleton and different components of the nuclear membrane and lamina (Tajik et al., 2016). Mechanical forces have also been proposed to repress gene transcription on longer timescales by inducing active chromatin condensation (Heo et al., 2016b) and the formation of a perinuclear cable of F-actin that depletes monomeric actin concentration in the nucleus (Le et al., 2016; Miroshnikova et al., 2017). Moreover, repositioning of perinuclear actin-rich domains and associated chromatin by mechanical forces has been suggested to modify the positioning of

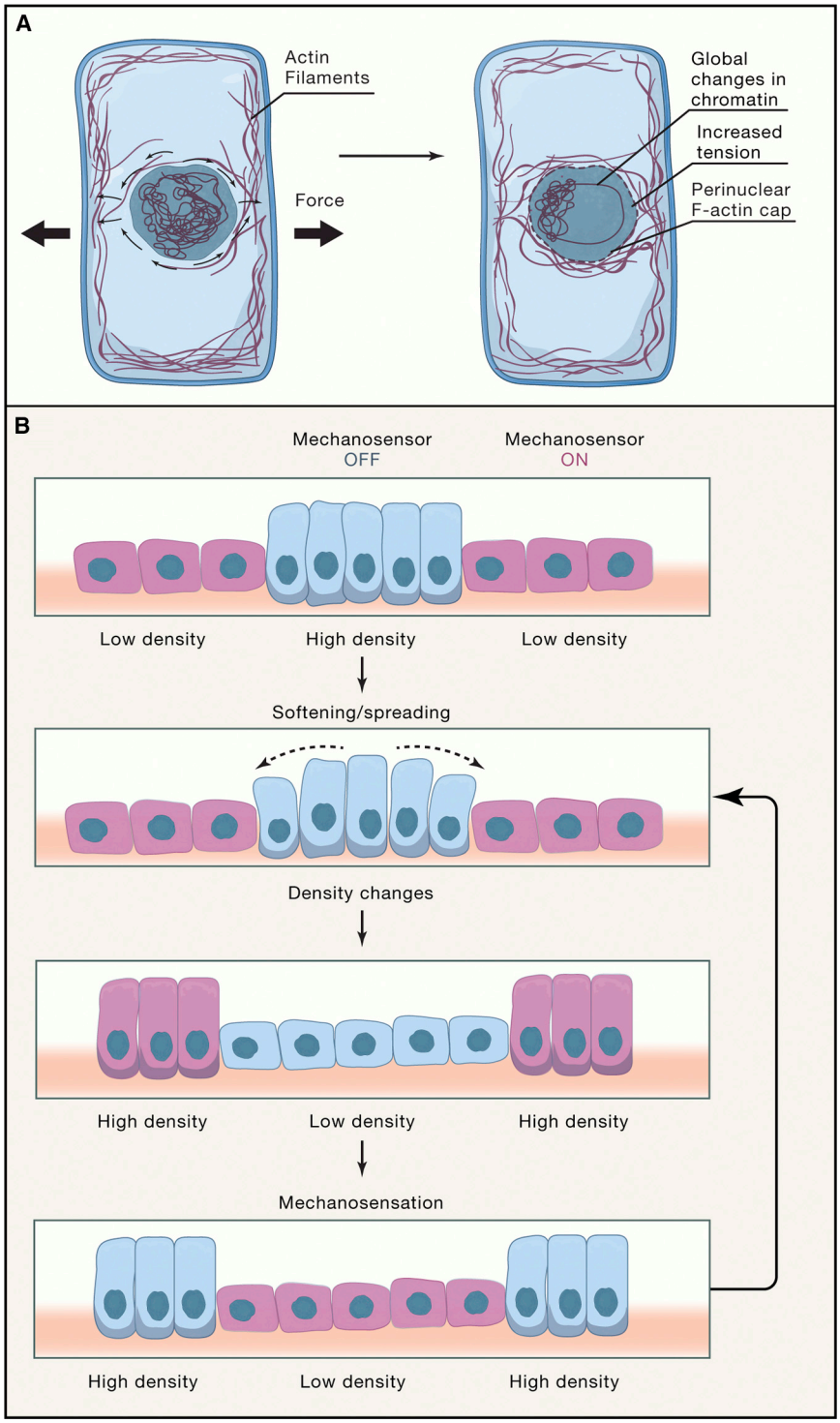


Figure 4. Mechanochemical Feedback Loops via Cell and Tissue Rheology

Mechanical forces can feed back on biochemical process via global rheological changes.

(A) At the cellular level, the nucleus is a mechanosensitive structure. Application of external forces, transmitted by the actomyosin cytoskeleton, can cause multiple global responses, including the formation of a perinuclear F-actin cap, which depletes G-actin from the nucleus, and chromatin stretching or condensation, causing global changes in gene transcription and changes in nuclear lamina tension that lead to its strengthening and/or microrupture (Cho et al., 2017).

(B) At the tissue level, spatiotemporal patterns of mechanosensor activity (for instance, ERK activity; high in purple, low in light blue) depend on local stress and density patterns but can also cause corresponding changes in cell migration, rheology, contractility, and/or spreading. Cellular responses, such as collective migration or nonlinear changes in cell rheology in turn can trigger changes in local cell density and mechanical stress (Aoki et al., 2017), feeding back on the mechanosensor activity that drives complex spatiotemporal dynamics, such as patterning and wave propagation.

chromosomal domains, thereby influencing the overall transcriptional landscape (Maharana et al., 2016; Miroshnikova et al., 2017). Collectively, these findings suggest that mechanical forces regulate global gene expression patterns by modulating chromatin organization within the nucleus (Figure 4A), potentially constituting a core module of mechanochemical feedback loops linking local mechanical stresses to global gene regulatory changes. Further work will be needed to understand how nuclear mechanosensing mechanistically controls chromatin packing and overall promoter accessibility in order to control the spatiotemporal pattern of gene expression *in vivo*. As chromatin packing, lamin expression, and nuclear rheology are all dynamically regulated during cell differentiation, mechanosensory changes in gene transcription within the nucleus are likely to feed back on the ability of the nucleus to act as a sensor and transducer of mechanical forces (Cho et al., 2017; Miroshnikova et al., 2017). This suggests new feedback loops where nuclear mechanics tune cellular differentiation patterns and vice versa. Recent theoretical efforts using realistic polymer physics models for chromatin (Bianco et al., 2018; Di Pierro et al., 2016; Giorgetti et al., 2014) are also likely to provide essential insights into understanding its reorganization in response to mechanical forces.

Changes in cell volume have also been proposed to function as a mechanosensory effector process mediating the effect of substrate stiffness on cell fate specification (Guo et al., 2017). While the exact role of cell volume changes in mechanosensation is still debated, cell volume changes could affect intracellular signal transduction pathways by changing global

rheological cell properties, such as cytoplasmic crowding. Indeed, modifying the fluid-protein ratio in a cell is predicted to generically affect any signaling cascade by changing the diffusion properties of their effector proteins, especially when crowding is so high that the system is close to a jamming transition (Miermont et al., 2013). This prediction is supported by experimental observations in yeast and human cell lines, where specific stress-response pathways, such as mTor signaling, are tuned by such intracellular crowding mechanisms (Delarue et al., 2018). Given the potential role of cell volume regulation in mechanosensing, one could thus envision a generic feedback loop between cell fate and mechanics mediated by global changes in cell rheological properties, such as intracellular crowding. This feedback could be mediated not only by changes in gene expression, but also in protein activity (see below). Recent advances in deciphering how the proteome and phosphoproteome changes upon force application during development (Hashimoto et al., 2019) are likely to shed light on this topic.

At the tissue level, clear evidence has emerged for several key cellular processes, such as cell migration and differentiation being mechanosensitive to the material properties of their environment, suggesting that mechanochemical feedback loops mediated by tissue rheological changes may play an important role during tissue morphogenesis. The MAPK-ERK pathway, in particular, has emerged as a central candidate integrating mechanical cues and cellular fate during development. Indeed, ERK signaling not only is mechanosensitive (Hirata et al., 2015; Schiller and Fässler, 2013), but also has been involved in modulating the activity and rheology of the actomyosin cytoskeleton (Aoki et al., 2017). Furthermore, the ERK-MAPK pathway is a central regulator of cell proliferation and differentiation (Kolch, 2005), with ERK inhibition being a defining feature for maintaining pluripotency in embryonic stem cells (Nichols and Smith, 2011). Studies on collective migration in epithelial cell sheets *in vitro* have provided some initial insight in the generation and function of mechanochemical feedback loops involving complex MAPK-ERK activity patterns (Aoki et al., 2017). They show that these patterns are closely correlated with the spontaneously emerging spatiotemporal patterns of cell density and polarity (Angelini et al., 2011; Aoki et al., 2017; Serra-Picamal et al., 2012). Interestingly, this correlation has been proposed to rely on multifaceted couplings between ERK activity, local cellular stresses, and cellular rheology, where mechanical stresses cause linear or non-linear changes in intracellular viscoelasticity (Aoki et al., 2017; Serra-Picamal et al., 2012) and cell polarization (Aoki et al., 2017; Banerjee et al., 2015), which feed back on ERK activity via mechanosensitive processes (Figure 4B). These mechanochemical feedback loops have been theoretically predicted to constitute robust routes toward complex spatiotemporal patterning instabilities, a prediction that still needs to be systematically addressed by experiments. Interestingly, similar patterns have recently been observed *in vivo*, such as in the mouse skin, where ERK signaling waves are strongly enhanced during wound healing (Hiratsuka et al., 2015). Furthermore, in *Drosophila* tracheal placode formation, invagination relies on a mutual feedback between a propagating wave of ERK activation and myosin II-dependent supracellular actomyosin cable formation (Ogura et al., 2018). How such mechanochemical pulses and waves of signaling pathway activation translate into cellular responses has only began to be elucidated (Tao et al., 2019), and the general topic of how signaling dynamics influence cellular fate outcomes, such as death, proliferation, or senescence, is a new and exciting research field.

Finally, although substrate rheology is well known to direct and modify cell migration from both *in vitro* studies and *in vivo* settings, the converse feedback of cell migration on substrate rheology has been much less studied. A potential example for such feedback loops has recently been proposed to underlie neural crest migration in *Xenopus* and zebrafish embryos (Barriga et al., 2018). Collective neural crest cell migration is triggered by a stiffening of the underlying head mesoderm, which itself arises from cell density increases caused by gastrulation movements. Likewise, abrupt rheological tissue transitions have most recently been proposed to play key roles for cell movements and tissue morphogenesis during zebrafish embryogenesis (Mongera et al., 2018; Petridou et al., 2019), and cell movements are likely to feed back to tissue rheology. Interestingly, ECM remodeling is a key defining feature of tumorigenesis, with increased ECM rigidity being correlated with cancer progression (Levental et al., 2009; Northcott et al., 2018). As ECM rheology has been shown to feed back on cancer cells in several ways, such as promoting epithelial-to-mesenchymal transition (EMT) (Wei et al., 2015), this suggests that mechanochemical feedback loop mediated by changes in tissue and/or substrate rheology might be shared between developmental and pathological processes.

Outlook

Mechanochemical feedbacks describe the bidirectional interactions between mechanical and chemical signals, which play key roles for morphogenesis and pattern formation in both developmental and disease-related processes. An inherent feature of such feedbacks is that they are intrinsically self-organizing, yielding to spontaneous patterning instabilities, which are not strictly dependent on polarizing external cues. Mechanochemical feedbacks are not the only theoretical solution to generate an organism: this could also arise either in a highly stereotypical, deterministic, and cell intrinsic manner, as described by classical embryology experiments, or via purely biochemical feedbacks (Turing, 1952), where mechanics would simply be a downstream effector of morphogenetic patterning. However, the recently reported prevalence of more complex interactions between biochemistry and mechanics suggests that they confer advantages for robust and modular morphogenesis (Gross et al., 2018). Still, self-organized feedback loops can also be risky, and their tight control is critical to avoid ectopic activation of key signaling pathways that would lead to developmental defects. As discussed in this Review, this is typically achieved by some polarizing external cues—for instance, the point where the sperm fertilizes the oocyte—or tissue geometry inherited from previous morphogenetic events, controlling the onset and biasing the position of self-organized patterns. The feedback as such is then needed to amplify this initial weak polarizing cue into robust cell and embryo polarization. This has led to broader theoretical questions on whether biological systems actively position themselves close to phase transitions to make

use of the enhanced sensitivity they confer (Krotov et al., 2014). Moreover, this allows the system to spontaneously polarize in case the initial polarizing cue is too weak or absent. For instance, the formation of organoids by stem cells might be initiated by the spontaneous activation of mechanochemical feedback loops that sets off a series of interdependent patterning and morphogenesis processes, eventually leading to organ formation (Ishihara and Tanaka, 2018; Serra et al., 2019). Moreover, in settings where a large number of patterns must be specified, such as the formation of hair follicles or feathers, intestinal villi, and animal coats, it is likely that—although a correct spacing is paramount—the precision in positioning of every single structure is not, suggesting that unguided self-organization might be best for the task.

Often—and this has been one of the main focus of this Review—the interaction between mechanical and chemical components in feedback loops involving cell fate specification and differentiation is much more indirect and thus less amenable for experimental and theoretical analysis. Typically, there is correlative evidence linking changes in cell and tissue properties, such as their geometry and rheology, to the spatiotemporal pattern of cell fate specification and differentiation. However, the feedback in such cases is highly indirect, involving long time delays associated with cell specification and differentiation, which hinder a definite correlation between the cause (e.g., changes in gene expression or protein activity) and effect (e.g., changes in cell mechanics modifying cell and tissue morphogenesis; Chan et al., 2017). In order to better understand such feedback loops, it will thus be imperative to elucidate how changes in gene transcription and cell signaling during cell fate specification affect the morphogenetic capacity of cells and tissues.

One potential way to experimentally tackle this challenge will be to improve live-imaging methods, allowing a more precise understanding of the temporal sequence of mechanical and biochemical events involved in morphogenesis. This could be done by further enhancing intravital live-imaging techniques, and/or improving uniform *ex vivo* preparations such as organs-on-a-chip, organoids or gastruloids (Beccari et al., 2018; Ishihara and Tanaka, 2018). An advantage of the latter approach is its amenability to controlled biochemical and biophysical perturbations, which are key to go beyond causality. Improvements of non-invasive methods to measure mechanical forces in tissues in order to achieve meaningful correlations between them and biochemical signaling activities will also be required. Another potential way will be to combine single-cell transcriptional trajectories, which provide unprecedented precision and insights into the process of cellular fate decisions but typically do not have detailed spatial information, with a multispectral imaging approach to identify cells *in situ*. This eventually should allow systematically correlating gene expression changes with the morphogenetic behavior of individual cells within the developing embryo. Once certain changes in gene expression can be correlated with a specific morphogenetic behavior, their causative relationship can be determined by conditional genetic perturbation experiments. With such knowledge in hand, one can then predict how gene expression changes triggered by mechanical signals feed back on those signals, thereby establishing a complete feedback loop between the mechanical signals determining tissue morphogenesis and the gene regulatory pathways involved in cell fate specification and differentiation. Finally, it will be important to develop new theoretical frameworks to generically integrate these couplings in a tractable-but-predictive manner, which will require the integration of multiple fields, from control or graph theory to out-of-equilibrium thermodynamics and biomechanics. Although different biological systems will likely require the development of different types of models, an important undertaking will be to systematically classify these models in order to understand whether generic classes of features arise across multiple systems.

Much progress has been made in recent years in unravelling the mechanical basis of cell and tissue morphogenesis and deciphering the gene regulatory pathways determining cell fate specification. It also becomes increasingly clear that these two processes are intertwined by mechanochemical feedback loops that coordinate embryo morphogenesis and patterning in both space and time. Future research will have to address how such mechanochemical feedback loops are realized on a molecular, cellular, and tissue level, and how they are controlled in development and disease. In particular, how those feedback loops are initiated by internal fluctuations versus external cues, how precise their outcome is for both embryo patterning and morphogenesis, and how these outcomes are stably maintained during adult homeostasis are important questions that need to be tackled by future efforts at the interface between biology and physics.

ACKNOWLEDGMENTS

We would like to thank Kevin Chalut, Stephan Grill, and Pierre Recho, as well as all the members of the Hannezo and Heisenberg groups for useful discussions and advice on the manuscript. We thank Claudia Flandoli for the artwork in the figures. We acknowledge funding from the European Union (European Research Council Advanced Grant 742573) to C.-P.H. and from the Austrian Science Fund (FWF) (P 31639) to E.H.

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